VULN-SWARM

Remediation Strategy Knowledge Base

RAG Reference Document — Document 2 of 2

Covers: Code Pattern Analysis · Dependency Audit · Secret Detection

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Purpose of This Document

This document is the Agent B (Engineer) primary RAG reference for remediation strategy. It contains research-backed, non-generic patching approaches for the three highest-signal vulnerability classes: Code Pattern Analysis, Dependency Audit, and Secret Detection.

Agent B must query this document before generating any patch. Agent C queries it to understand what a correctly patched version should look like so it can write accurate verification tests.

Every strategy in this document is derived from published security research, production tools, or established standards (NIST, OWASP, MITRE, SLSA). Generic advice is explicitly excluded.

Part A: Code Pattern Analysis — Remediation Strategies

Code pattern vulnerabilities require context-aware fixes, not simple substitutions. The core principle across all strategies is the Principle of Least Authority (POLA): grant the minimum capability the code needs to function.

A.1 SQL Injection — Taint Sink Mapping Strategy

Vuln ID	CP-001
Research Basis	CodeQL taint tracking · Facebook Pysa (Python Static Analyser) · CWE-89 canonical remediation
Core Principle	Fix at the last safe transformation point before the sink — not at the input source, not naively at the sink.

The Taint Sink Mapping Approach

Agent B must identify the full taint flow from source to sink using the Data Flow Graph before deciding where to apply the fix. The naive approach of sanitising at the input source is wrong because intermediate transformations can re-introduce the tainted value.

```
WRONG: Sanitise at source (too broad, misses re-taint paths)
def api_endpoint(user_input):
    clean = sanitise(user_input)   # fix applied here
    result = process(clean)         # process() may re-introduce taint
    query = build_query(result)
    db.execute(query)               # sink still receives tainted data
```

```
CORRECT: Parameterise at the sink (the only reliable fix point)
def api_endpoint(user_input):
    result = process(user_input)
    query = build_query(result)
    db.execute(query, params=[result])   # parameterised at sink
```

Dynamic Identifier Fix (Tables / Columns)

Parameterised queries cannot parameterise table or column names. When the vulnerable code uses dynamic identifiers, Agent B must apply allowlist-based identifier validation.

```
Dynamic table name — cannot be parameterised
ALLOWED_TABLES = {'users', 'products', 'orders'}
if table_name not in ALLOWED_TABLES:
    raise SecurityError(f'Invalid table: {table_name}')
query = f'SELECT * FROM {table_name}'  # safe: identity verified by allowlist
```

ORM Raw() Escape Hatch Fix

When Agent A detects use of ORM raw() methods with string formatting, Agent B must decompose the query back into the ORM's safe query builder.

```
Unsafe: bypasses ORM parameterisation
User.objects.raw(f"SELECT * FROM users WHERE id={user_id}")
```

```
Correct: ORM handles parameterisation
```

```
User.objects.filter(id=user_id)
```

A.2 SSTI — SandboxedEnvironment + Template Complexity Limiting

Vuln ID	CP-003
Research Basis	PortSwigger SSTI Research (2015) · Jinja2 SandboxedEnvironment documentation · OWASP SSTI prevention cheatsheet
Core Principle	Never substitute with a 'safer' template call — sandbox the entire template execution environment.

SandboxedEnvironment Replacement

The fix is not to replace Template() with a different call. The fix is to replace the entire template execution environment with one that explicitly restricts capability.

```
Dangerous: user controls the template
from jinja2 import Template
template = Template(user_input)
return template.render()
```

```
Correct: SandboxedEnvironment with explicit restriction
from jinja2.sandbox import SandboxedEnvironment
env = SandboxedEnvironment()
env.globals = {}       # no globals exposed to template
env.filters = {}       # no filters exposed to template
template = env.from_string(user_input)
return template.render(safe_context)  # only pass safe, typed context
```

Template Complexity Limiter

Even a sandboxed environment can be DoS'd with deeply nested or extremely complex templates. Agent B adds a complexity limiter that rejects templates exceeding a safe AST depth.

```
import jinja2
def validate_template_complexity(template_str: str, max_depth: int = 5):
    env = jinja2.Environment()
    ast = env.parse(template_str)
    depth = _measure_ast_depth(ast)
    if depth > max_depth:
        raise SecurityError('Template complexity exceeds safe limit')
```

A.3 Path Traversal — Realpath Jail Validation

Vuln ID	CP-004
Research Basis	CWE-22 canonical remediation (MITRE) · OWASP path traversal prevention guide
Core Principle	os.path.join() alone is NOT sufficient. Symlinks can still escape the directory. Realpath resolution is the only traversal-proof approach.

Why os.path.join() Fails

The common fix of using os.path.join() still allows path traversal because it does not resolve symlinks before boundary checking.

```
Still vulnerable despite os.path.join
base = '/var/app/files'
path = os.path.join(base, '../../etc/passwd')
# Result: /etc/passwd  — traversal succeeded
```

Realpath Jail — The Only Correct Fix

Agent B must replace all path construction logic with the realpath jail pattern. This resolves ALL symlinks and relative components before performing boundary validation.

```
import os

def safe_file_access(base_dir: str, user_path: str) -> str:
    # Resolve ALL symlinks and relative components first
    resolved = os.path.realpath(
        os.path.join(base_dir, user_path)
    )
    # Validate boundary AFTER full resolution
    jail = os.path.realpath(base_dir)
    if not resolved.startswith(jail + os.sep):
        raise SecurityError('Path traversal detected')
    return resolved
```

A.4 IDOR — Security Invariant Injection

Vuln ID	CP-007
Research Basis	OWASP ASVS Level 2 · Security Policy as Code pattern · NIST SP 800-218 Section 2.3
Core Principle	Do not patch 47 routes individually. Enforce ownership invariants at the framework level so all current and future routes inherit the fix.

Invariant Injection at Framework Level

Agent B identifies the routing registration pattern and injects security invariants as middleware or decorators at the application level, not at each endpoint.

```
Invariant: every route with a user_id param must verify ownership
@app.before_request
def enforce_ownership_invariant():
    user_id = request.view_args.get('user_id')
    if user_id and str(request.current_user.id) != str(user_id):
        raise Forbidden('Ownership check failed')
```

Agent B then patches ONLY the explicit exceptions to this invariant (admin routes, public routes), rather than patching every protected endpoint individually.

A.5 Missing Rate Limiting — Endpoint-Aware Limiter

Vuln ID	CP-008
Research Basis	OWASP API Security Top 10 · NIST SP 800-63B (brute force protections)
Core Principle	Rate limits must be differentiated by endpoint sensitivity — login gets 5/minute, public search gets 100/minute. Blanket limits break functionality.

```
from slowapi import Limiter
from slowapi.util import get_remote_address
limiter = Limiter(key_func=get_remote_address)

Authentication endpoints: strict limit
@app.route('/login', methods=['POST'])
@limiter.limit('5/minute')
def login(): ...

Password reset: even stricter
@app.route('/reset-password', methods=['POST'])
@limiter.limit('3/hour')
def reset_password(): ...
```

Part B: Dependency Audit — Remediation Strategies

Dependency vulnerabilities require three separate strategies that most teams collapse into one (just upgrade). Agent B must apply all three independently.

B.1 Reachability Analysis — Before Any Upgrade

Applies To	DEP-001, DEP-002
Research Basis	Endor Labs Reachability Research (2023) · Google OSV-Scanner v2 whitepaper · OWASP Dependency Track
Core Principle	Not all CVEs in a dependency affect your code. Only patch when the vulnerable function is provably reachable from your codebase.

The Reachability Algorithm

Agent B runs this algorithm before deciding on any dependency upgrade. The output determines whether the CVE is a confirmed critical finding or a present-but-unexploitable finding.

1. Extract the specific vulnerable function/class from the CVE advisory.
2. Build a call graph of the entire codebase from all entry points (routes, main, celery tasks, CLI commands).
3. BFS through the call graph: does any execution path reach the vulnerable library function?
4. UNREACHABLE: mark as low priority in PatchManifest. Upgrade is still recommended but not blocking.
5. REACHABLE: confirmed critical. Full patch pipeline. Upgrade is mandatory before PR merge.

```
Reachability check pseudo-implementation for Agent B
def is_vuln_reachable(codebase_path, vuln_function: str) -> bool:
    call_graph = build_call_graph(codebase_path)
    entry_points = find_entry_points(codebase_path)
    for entry in entry_points:
        if bfs_reaches(call_graph, entry, vuln_function):
            return True
    return False
```

B.2 Semantic Compatibility Graph — Safe Upgrade Path

Applies To	DEP-001, DEP-002, DEP-003
Research Basis	Dependabot compatibility scoring methodology · Semantic versioning specification · librariesio API
Core Principle	Before upgrading, compute exactly which public API methods your code uses and verify none are removed or changed in the target version.

Safe Upgrade Path Computation

6. Static import analysis: find every requests.*, flask.*, etc. call your codebase makes.
7. API diff: compare public API surface of current version vs target version using library changelogs and signature inspection.

8. Intersection: do any of YOUR used methods appear in the CHANGED or REMOVED set?
9. No intersection: safe to upgrade. Bump version only.
10. Intersection found: Agent B must patch the affected call sites in the codebase AND bump the version in the same commit.

B.3 Transitive Dependency Materialisation

Applies To	DEP-002
Research Basis	SLSA Supply Chain Threats taxonomy Level 3 · Google Open Source Insights · pip-audit transitive mode
Core Principle	requirements.txt only shows top-level deps. The vulnerability is almost always in a dep-of-a-dep. Full tree materialisation is mandatory.

Full Tree Materialisation Command

```
pip install pipdeptree
pipdeptree --json-tree > full_dep_tree.json
# CVE scan every node in the tree, not just top-level packages
```

Fix Strategy for Transitive Vulnerabilities

Agent B chooses between two strategies based on the semantic compatibility check:

- Strategy A — Fix at parent: upgrade the direct dep to a version that pulls in a safe transitive dep. Cleaner solution, fewer lines changed.
- Strategy B — Pin directly: add the transitive dep explicitly to requirements.txt with a safe version. Used when upgrading the parent would break the codebase.

B.4 Hash Pinning — Beyond Version Pinning

Applies To	DEP-003, DEP-004
Research Basis	PEP 508 hash-checking mode · pip-tools pip-compile · SLSA Level 2 provenance requirements
Core Principle	Version pinning (==2.31.0) is insufficient. The package at that version can be replaced on PyPI. SHA-256 hash pinning prevents substitution attacks entirely.

Hash-Pinned requirements.txt Format

Agent B generates hash-pinned entries by fetching the package wheel and sdist, computing their SHA-256 digests, and embedding both.

```
Agent B produces this format for all production dependencies
requests==2.31.0 \
    --hash=sha256:942c5a758f98...  \   # wheel hash
    --hash=sha256:58cd2187423...       # sdist hash
```

Both hashes are required: the wheel hash covers binary installs and the sdist hash covers source installs. A single hash is insufficient.

Part C: Secret Detection — Remediation Strategies

Secret remediation is a two-phase problem: detection accuracy and complete remediation. Most teams only solve phase one. Agent B must solve both.

C.1 Entropy + Context Fusion Detection

Applies To	SEC-001, SEC-002
Research Basis	GitGuardian detection engine architecture · Trufflehog v3 multi-pass scanner · Shannon entropy theory applied to credential detection
Core Principle	Regex alone has a 40-60% false positive rate. Entropy scoring combined with context weighting reduces this to under 5%.

The Fusion Scoring Formula

Agent A computes a combined score. Only values exceeding the threshold are escalated.

```
score = (shannon_entropy_score)
      * (variable_name_proximity_score)
      * (file_context_score)
      * (pattern_match_score)
```

Shannon Entropy	Strings with >4.5 bits/char are likely real secrets. Lower entropy indicates placeholder text.
Variable Proximity	String assigned to variable containing: key, token, secret, password, api, auth, credential = higher score.
File Context	In test files: score reduced. In config, settings, .env, or deployment files: score increased.
Pattern Match	Known service prefixes (sk-ant-, ghp_, AKIA, xoxb-) apply the highest multiplier.

C.2 Multi-Pass Obfuscation Detection

Applies To	SEC-001, SEC-002
Research Basis	Trufflehog v3 open-source scanner architecture · GitGuardian obfuscation research
Core Principle	AI agents split, encode, or obfuscate secrets to avoid naive scanners. Multi-pass scanning reconstructs these before applying entropy scoring.

Agent A runs four passes in sequence:

11. Direct scan: entropy + pattern matching on all string literals.
12. Concatenation reconstruction: trace all string + operations and variable assignments to reconstruct composite strings, then rescan.
13. Encoding decode: detect and decode base64, hex, and rot13 encoded strings, then rescan the decoded value.
14. Cross-reference: compare detected secrets against known-leaked credential databases (HaveIBeenPwned API, GitHub secret scanning DB).

```
Obfuscation example Agent A must catch
key_part1 = 'sk-ant-api'
key_part2 = '03-realSecretHere'
api_key = key_part1 + key_part2   # reconstructed in Pass 2
```

```
import base64
api_key = base64.b64decode('c2stYW50...')  # decoded in Pass 3
```

C.3 Complete Remediation — Beyond Moving to .env

Applies To	SEC-001, SEC-002, SEC-003
Research Basis	NIST SP 800-218 Section 2.5 · GitHub Secret Scanning Advisory · git filter-repo documentation
Core Principle	Moving a secret to .env does NOT fix the vulnerability if it was ever committed. The secret is in git history and is fully accessible. Rotation is mandatory.

Agent B's Two-Phase Secret Fix

Phase 1 — Code Fix (Agent B executes):

15. Replace hardcoded secret with environment variable reference in current code.
16. Add .env to .gitignore if not already present.
17. Add .env.example with placeholder value documenting the required variable.
18. Flag in PatchManifest: HISTORY_PURGE_REQUIRED = True.

Phase 2 — Human Actions (Agent C raises GitHub Issue):

19. Rotate the secret immediately — generate a new token. The old one is compromised regardless of the code fix.
20. Run git filter-repo to purge the secret from all commits in history.
21. Force push all branches after history rewrite.
22. Notify all contributors to re-clone the repository.
23. Revoke the old token at the service provider (GitHub, AWS, Anthropic, etc.).

Git History Purge Command (in GitHub Issue)

```
# Install git-filter-repo
pip install git-filter-repo

# Create expressions file
echo 'regex:sk-ant-[a-zA-Z0-9-_]+==>REDACTED' > expressions.txt

# Purge from all history
git filter-repo --replace-text expressions.txt

# Force push all branches
git push --force --all
```

C.4 Dead Secret Handling

Applies To	SEC-001, SEC-002
Research Basis	NIST SP 800-57 credential lifecycle management
Core Principle	An expired or rotated secret still in code is a security finding. It proves the team committed live secrets historically and reveals which services are in use.

Agent A validates liveness before severity classification:

- ACTIVE secret (live API call returns 200): CRITICAL severity. Immediate full pipeline.
- DEAD secret (API returns 401/403): MEDIUM severity. Still goes through pipeline. GitHub Issue notes the finding for audit purposes and still requires history purge.

Agent Query Reference — This Document

Agent B — Query Patterns

- Query: 'how to fix SQL injection taint sink' — returns Part A.1 sink mapping strategy
- Query: 'path traversal realpath fix' — returns Part A.3 jail validation code
- Query: 'dependency upgrade safe steps' — returns Part B.2 compatibility graph steps
- Query: 'transitive vulnerability fix strategy A vs B' — returns Part B.3 decision criteria
- Query: 'secret in git history what to do' — returns Part C.3 two-phase remediation

Agent C — Query Patterns

- Query: 'what does correct SSTI patch look like' — returns Part A.2 SandboxedEnvironment pattern
- Query: 'how to verify hash pinning is applied' — returns Part B.4 requirements.txt format
- Query: 'dead secret severity classification' — returns Part C.4 liveness validation criteria
- Query: 'IDOR invariant fix verification' — returns Part A.4 middleware pattern to test against

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Failure Taxonomy of Autonomous Remediation: Analyzing the Remediation Blindspots of LLM-Based Security Swarms

The rapid integration of large language models (LLMs) into critical software infrastructure has fundamentally shifted the cybersecurity landscape from isolated algorithmic errors toward complex systemic vulnerabilities.[1] As engineering organizations deploy autonomous agentic architectures to detect and remediate software flaws, they encounter a new class of emergent risks. In a multi-agent vulnerability remediation swarm, the architecture typically relies on a distinct division of labor: a reconnaissance and exploitation agent (Agent A) that identifies flaws, a remediation agent (Agent B) tasked with synthesizing a patch, and a verification agent (Agent C) that audits the proposed fix.[2]

However, empirical evidence demonstrates that Agent B, functioning as an LLM-driven software engineer, predictably introduces secondary vulnerabilities, functional regressions, and incomplete mitigations during the patching process.[4] This phenomenon is driven by the "Format-Reliability Gap"—a mechanistic failure where the model's security representations, while encoded in the earliest computational layers, remain computationally inert until the final layers, where they are overwhelmed by the immediate demand to produce format-compliant, functionally executing code.[6] Consequently, the LLM prioritizes surface-level syntactic fixes over structural security guarantees, resulting in what is known as "architectural drift," where subtle, model-generated design changes break security invariants without violating syntax.[7]

The systemic failures of autonomous agents rarely arise from isolated technical defects; instead, they emerge from recurring combinations of design shortcomings and a fundamental misunderstanding of the adversarial execution environment.[1] Traditional software development requires an understanding of context—such as network boundaries, the isolation level of database transactions, or the execution parameters of a continuous integration pipeline. LLMs, despite their vast training corpora, are localized sequence predictors that lack a holistic architectural mental model. When instructed to fix a vulnerability quickly, Agent B defaults to the most statistically probable solution found in its training data, which often consists of rapid prototyping examples, insecure StackOverflow answers, or non-production-ready development configurations.[8]

This report provides a comprehensive, highly technical taxonomy of the specific mistakes, anti-patterns, and naive patches generated by autonomous remediation agents across six critical vulnerability classes. Furthermore, it defines the rigorous static and dynamic verification matrices required by the auditing agent (Agent C) to detect, analyze, and reject these flawed

remediations. The analysis draws upon state-of-the-art vulnerability data, specific Common Vulnerabilities and Exposures (CVEs), and architectural paradigms to construct an exhaustive framework for securing agentic remediation pipelines.

1. Secret Leaks and Environment Mismanagement

The Vulnerability & The Exploit Context

During an automated source code review or dynamic reconnaissance phase, Agent A identifies that an application is utilizing hardcoded, legacy API tokens, exposing database credentials, or failing to enforce authentication on a critical internal microservice. Agent A successfully exploits this by directly calling the exposed endpoint or extracting the hardcoded string from the source repository, thereby exfiltrating sensitive data. The context provided to Agent B indicates an urgent, high-severity requirement to secure the endpoint, remove the hardcoded secrets from the repository, and implement a secure configuration management strategy.

Agent B's "Naive Patch" (The Mistake)

To eliminate the hardcoded secret, Agent B generates a patch that dynamically loads the secret from an environment configuration file. Because LLMs are heavily biased toward modern JavaScript and Python development paradigms, the model frequently leverages popular third-party libraries designed for local development to parse configuration files. In a Python context, the model will predictably deploy the dotenv library to orchestrate this configuration change.

Python

```python
# Agent B's Flawed Remediation Patch
import os
from dotenv import load_dotenv
import requests

def send_secure_message(payload):
    # The LLM implicitly loads the entire environment, including hidden.env files
    load_dotenv(override=True)

    api_key = os.getenv("CRITICAL_API_KEY")
    if not api_key:
        raise ValueError("API Key not found in configuration")

    headers = {"Authorization": f"Bearer {api_key}"}
    return requests.post("https://api.internal/v1/message", headers=headers, json=payload)
```

Simultaneously, to satisfy the requirement for rigorous testing and continuous integration (CI) compliance, Agent B generates a unit test script that validates the presence of the newly abstracted environment variable:

Python

```python
# Agent B's Flawed CI/CD Test Script
import os

def test_send_secure_message_config():
    # Naive test generation that echoes the secret to CI/CD logs
    api_key = os.getenv("CRITICAL_API_KEY")
    print(f"DEBUG: Initializing test sequence. Using API Key {api_key} for testing")
    assert api_key is not None
```

The Fallacy

Agent B's logic exhibits two distinct cognitive blindspots that transform a static code vulnerability into a dynamic runtime and pipeline exploit. First, by implementing the load_dotenv() function, the LLM introduces a mechanism that automatically searches the local filesystem for .env, .env.local, and similar configuration files, loading their contents directly into the application's memory space and the host environment.[9] The model assumes that because environment variables are not hardcoded in the source code, the system is inherently secure. It fails to recognize that development utilities silently load these secrets into memory, where they become accessible to other processes, malicious dependencies, or prompt-injection payloads running within the same execution context.[9] The Anthropic Claude Code agent incident highlights this exact paradigm, where an agent automatically parsed .env files from a user's directory and loaded HTTP proxy credentials into memory without explicit authorization, resulting in silent credential exposure.[9]

Second, the model lacks an understanding of the adversarial implications of Continuous Integration/Continuous Deployment (CI/CD) environments. By generating a test case that prints the API key for debugging purposes, Agent B inadvertently creates a pipeline exploit.[10] Malicious or naive test code generated by an LLM can easily access environment variables and exfiltrate secrets to external servers or expose them in publicly accessible build logs.[11] The LLM optimizes purely for test coverage and functional visibility, entirely ignoring the operational security of the CI/CD pipeline, thereby substituting a static code leak for an operational logging leak.[10]

Agent C Verification Matrix

To prevent the introduction of environmental mismanagement and CI/CD secret exposure, Agent C must implement a rigorous verification matrix that evaluates both the syntactic structure of the patch and its runtime behavior within an instrumented sandbox.

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria
Static Analysis (AST)	Traverse the Abstract Syntax Tree (AST) for CallExpression nodes invoking .env parsing libraries (e.g., load_dotenv(), config()) in production code paths.	Flag the usage of local environment loaders outside of local development environments. The patch must rely on native OS-level environment injection provided by the container orchestration platform.
Static Analysis (AST)	Execute inter-procedural data-flow taint tracking, tracing the outputs of os.getenv(), process.env, or System.getenv() to CallExpression nodes representing standard output sinks (e.g., print(), console.log(), logging.debug()).	Reject the patch if any tainted secret variable reaches an unencrypted logging sink, standard output, or test assertion logging mechanism.[11]
Dynamic Validation	Execute the proposed patch within an ephemeral, restricted CI sandbox. Inject a recognizable, high-entropy honey-token environment variable (e.g., sk-live-HONEYTOKEN12345) prior to test execution.	Monitor the standard output streams, system logs, and outgoing network requests via an egress proxy. If the honey-token appears in the console logs or is transmitted to an unapproved origin, the patch is highly destructive and must be rejected.[11]

Dynamic Validation	Monitor the filesystem interactions of the patched application utilizing system call tracing (e.g., strace or ptrace).	If the application actively probes for or opens .env files in production execution modes, Agent C must flag the application for insecure environment management practices.[9]

2. Insecure Configurations: CORS and Docker

The Vulnerability & The Exploit Context

Agent A identifies two separate architectural misconfigurations within a modern microservices deployment. First, a web application API is vulnerable to Cross-Origin Resource Sharing (CORS) abuse due to a missing Access-Control-Allow-Origin header. This misconfiguration prevents legitimate cross-origin frontend applications from fetching required resources. Second, a containerized workload fails to execute a required system diagnostic command, returning a strict Permission Denied error during runtime execution.[12]

Agent B's "Naive Patch" (The Mistake)

To resolve the CORS accessibility issue, Agent B defaults to the most permissive configuration available to ensure functional connectivity. In a Node.js backend utilizing the Express framework, the LLM will reliably generate the following initialization code:

JavaScript

```javascript
// Agent B's Flawed CORS Patch
const express = require('express');
const cors = require('cors');
const app = express();

// The LLM deploys a wildcard origin policy while explicitly enabling credentials
app.use(cors({
  origin: '*',
  credentials: true
}));

app.get('/api/user/profile', requireAuth, async (req, res) => {
  const user = await User.findById(req.user.id);
  res.json(user);
```

```
});
```

To resolve the Docker permission issue and ensure the diagnostic tool executes successfully, Agent B applies a blunt escalation of privileges to the container runtime, modifying the docker-compose.yml or container launch arguments:

YAML

```yaml
# Agent B's Flawed Docker Privileged Patch
services:
  backend-service:
    image: internal-api:latest
    # The LLM escalates to privileged mode to bypass permission errors
    privileged: true
    user: "0:0"
    network_mode: "host"
    volumes:
      # The LLM exposes the Docker daemon to the container
      - /var/run/docker.sock:/var/run/docker.sock
```

The Fallacy

In the context of the Cross-Origin Resource Sharing (CORS) remediation, Agent B optimizes for functional interoperability over the principle of least privilege. The LLM understands that the client application is experiencing a cross-origin blocking error and immediately deploys the * wildcard to instruct the browser to bypass the Same-Origin Policy (SOP).[13] However, the LLM suffers from a severe architectural blindspot: the HTTP specification expressly forbids the use of the * wildcard when the Access-Control-Allow-Credentials flag is set to true.[14] Modern browsers will categorically reject this configuration. In response to this browser rejection, a more advanced (but equally flawed) LLM patch might dynamically reflect the Origin header from the request back to the client, effectively bypassing the browser's safeguard while functionally mimicking a wildcard. When an LLM configures the server to manually reflect the requested origin, it allows any arbitrary domain on the internet to execute authenticated cross-origin requests, leading to severe session hijacking, credential exposure, and state-changing exploitation.[13]

Regarding the Docker configuration, Agent B treats the Permission Denied error as a localized file-access or basic Linux user permission issue rather than a fundamental container isolation feature. By injecting privileged: true, modifying the user to root (0:0), altering the network namespace to host, or mounting the Docker socket (/var/run/docker.sock), the LLM essentially nullifies the container's security boundary.[17] The model fails to comprehend that the Docker

socket provides unmitigated root access to the host machine, allowing any process within the container to instantiate sibling containers with full host privileges.[18] Furthermore, the privileged flag disables crucial namespace isolation and seccomp filters. When a container runs in privileged mode, it exposes the Linux Devices cgroup, allowing the container to utilize the mknod system call to create and access character and block devices on the host kernel, such as /dev/sda.[19] Recent benchmark studies, such as SandboxEscapeBench, demonstrate that frontier models routinely exploit these exact misconfigurations—specifically writable host mounts and exposed Docker sockets—to achieve a 100% escape rate from containerized environments.[20] The LLM views these configuration directives merely as functional switches to ensure the software runs, entirely oblivious to the fact that it has provisioned a trivial, zero-day container escape vector.

Agent C Verification Matrix

Evaluating infrastructure-as-code (IaC) and middleware configurations requires Agent C to implement a hybrid approach, combining static YAML/JSON parsing with dynamic environmental probing.

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria
Static Analysis (AST)	Traverse the AST for CORS middleware configurations (e.g., Express cors(), Spring @CrossOrigin). Identify Property nodes where the key is origin and the value is a StringLiteral containing * or a function that blindly reflects the request's origin.[13]	Reject patches that combine a wildcard origin (or reflected origin) with an ObjectExpression enabling credentials: true. Require explicit, hardcoded arrays of permitted frontend domains.[13]
Static Analysis (AST)	Parse YAML structures for Docker, Kubernetes, or pod specifications. Locate keys matching privileged: true, user: "0:0", pid: host, network_mode: host, or volumes mapping /var/run/docker.sock.[18]	Reject any configuration that mounts the Docker socket, escalates to privileged mode, or modifies the PID namespace. Mandate the use of specific cap_add directives rather than blanket privilege escalation.[18]

Dynamic Validation	Dispatch a mock OPTIONS preflight HTTP request originating from a known untrusted domain (e.g., Origin: https://malicious-actor.com).	If the server responds with Access-Control-Allow-Origin: https://malicious-actor.com alongside Access-Control-Allow-Credentials: true, the patch is dynamically reflecting untrusted origins and must be rejected.[16]
Dynamic Validation	Deploy the container payload in a heavily monitored sandboxed runtime. Execute an automated container escape payload targeting the Linux Devices cgroup (attempting mknod) or attempting to query the mounted Docker socket.[19]	If the payload successfully reads from the host's /etc/shadow, mounts a host block device, or instantiates an adjacent container via the socket, the isolation boundary is broken and the patch fails.[20]

3. Server-Side Request Forgery (SSRF) and URL Validation

The Vulnerability & The Exploit Context

Agent A discovers that a web application allows users to supply a URL to fetch remote assets, such as user avatars, document rendering webhooks, or open-graph metadata generators. By supplying a crafted payload such as http://169.254.169.254/latest/meta-data/ or utilizing a DNS rebinding service, Agent A successfully executes a Server-Side Request Forgery (SSRF) attack. The application's backend server processes the request and returns highly sensitive cloud provider metadata credentials to the attacker.[23]

Agent B's "Naive Patch" (The Mistake)

Agent B attempts to remediate the SSRF vulnerability by implementing a regular expression (regex) to enforce a strict allowlist of permitted domains. A common, highly flawed pattern generated by LLMs in languages like Go (mirroring the real-world Fulcio CVE-2022-28946 vulnerability) involves dynamic regex compilation:

Go

```go
// Agent B's Flawed SSRF Validation Patch
package config

import (
    "regexp"
    "strings"
)

func metaRegex(issuer string) (*regexp.Regexp, error) {
    // The LLM quotes the issuer string and replaces wildcards to build a regex
    quoted := regexp.QuoteMeta(issuer)
    replaced := strings.ReplaceAll(quoted, regexp.QuoteMeta("*"), "[-_a-zA-Z0-9]+")

    // FATAL MISTAKE: Missing ^ and $ anchors during compilation
    return regexp.Compile(replaced)
}

func validateURL(target string) bool {
    regex, _ := metaRegex("https://oidc.eks.*.amazonaws.com/id/*")
    return regex.MatchString(target)
}
```

In Python applications, the LLM may introduce an alternative flaw related to URL parsing logic:

Python

```python
# Agent B's Flawed Python SSRF Parsing Patch
from urllib.parse import urlparse
import requests

def fetch_webhook(url):
    parsed = urlparse(url)
    if parsed.hostname!= "trusted.internal.service":
        raise ValueError("Invalid hostname")

    # FATAL MISTAKE: Parser desynchronization
    response = requests.get(url)
```

```
return response.content
```

The Fallacy

Agent B's implementation exposes a profound misunderstanding of string validation semantics across different programming languages and regular expression engines. In the Go language snippet, the regexp.MatchString() function performs *substring matching* by default. This means that if the compiled regular expression pattern is found anywhere within the target string, the function returns true.[25] Because the LLM failed to explicitly include the start-of-string (^) and end-of-string ($) anchors during the regex compilation phase, the validation is entirely unanchored.[25]

The LLM incorrectly assumes that evaluating a regular expression against a string enforces an exact match constraint. Consequently, an attacker can trivially bypass the patch by embedding the trusted domain within the path or query parameters of a malicious URL. If the attacker supplies the payload https://attacker.com/x/https://oidc.eks.foo.amazonaws.com/id/bar, the substring match succeeds because the trusted pattern exists within the attacker's string. The SSRF protection is completely bypassed, and the backend server is tricked into routing traffic to the attacker-controlled infrastructure.[25]

Furthermore, LLMs frequently fall victim to URL parsing inconsistencies across different software libraries. They may use one library (e.g., Python's urllib.parse or urllib3) to extract and validate the hostname, but rely on a completely different asynchronous or high-level library (e.g., requests, aiohttp, or yarl) to execute the actual HTTP request.[26] Because these distinct libraries handle malformed paths, unicode characters, percent-encoding tricks, and protocol smuggling differently, the validation layer and the execution layer fall out of synchronization. An attacker can construct a malformed URL that urllib.parse interprets as safe, but which the execution client interprets as a local loopback address. This specific mechanism was responsible for the CVE-2026-25960 vulnerability in the vLLM serving engine, where urllib3 was used for validation but aiohttp executed the request, allowing an attacker to smuggle an SSRF payload past the regex filter.[26]

Agent C Verification Matrix

Agent C must evaluate URL validation logic with extreme prejudice, as string manipulation is notoriously brittle against modern SSRF evasion techniques.

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria
Static Analysis (AST)	Inspect CallExpression nodes invoking regular expression compilation	Ensure that URL validation patterns explicitly begin with the ^ anchor and

	(e.g., regexp.Compile, new RegExp, re.compile). Analyze the StringLiteral patterns being passed as arguments.[25]	terminate with the $ anchor. Categorically reject any unanchored domain validation logic.[28]
Static Analysis (AST)	Conduct deep data-flow analysis to identify instances where the URL parser used for host extraction differs from the HTTP client executing the outbound network request.	Flag mismatched parsing libraries (e.g., validating the host with urllib.parse but executing the request with aiohttp or requests) to prevent parser desynchronization and protocol smuggling vulnerabilities.[26]
Dynamic Validation	Execute targeted SSRF payloads designed to exploit unanchored matching (e.g., http://malicious.com/a.html#http://trusted.com or http://trusted.com.malicious.com).[29]	If the HTTP client resolves the URL and executes a request to the malicious host, the regular expression validation is flawed and the patch is rejected.
Dynamic Validation	Execute DNS rebinding attacks and IP obfuscation payloads against the patched endpoint. Utilize payloads such as octal representations (http://0177.0.0.1), hexadecimal IPs, or IPv6 mapped IPv4 addresses.[27]	The patch must reject all payloads resolving to internal loopback addresses (127.0.0.1), private RFC1918 space, or cloud metadata IPs, regardless of the encoding mechanism or DNS resolution strategy utilized by the attacker.[27]

4. Insecure Direct Object Reference (IDOR)

The Vulnerability & The Exploit Context

Agent A manipulates the identification parameter in an API request payload or URL parameter (e.g., altering a GET request from /api/v2/users/123456/orders to /api/v2/users/123457/orders). The application processes the request and successfully returns, modifies, or deletes objects

belonging to another user. This constitutes an Insecure Direct Object Reference (IDOR), a severe failure in object-level authorization resulting in horizontal or vertical privilege escalation.[30]

Agent B's "Naive Patch" (The Mistake)

To fix the IDOR vulnerability, Agent B reviews the endpoint and identifies that while global authentication might be present, the specific database query lacks a constraint to verify ownership. In a modern TypeScript environment utilizing an Object-Relational Mapper (ORM) like Prisma or TypeORM, the LLM will reliably generate the following flawed patch:

TypeScript

```typescript
// Agent B's Flawed IDOR Remediation Patch
import { prisma } from '../lib/db';

app.patch('/api/folders/:id', requireAuth, async (req, res) => {
  const { id } = req.params;
  const { name } = req.body;

  // The LLM correctly authenticates the user but fails to scope the query
  // This query relies solely on the object ID for the update operation
  const updatedFolder = await prisma.folder.update({
    where: {
      id: Number(id) // Missing contextual ownership verification
    },
    data: { name }
  });

  res.json(updatedFolder);
});
```

Alternatively, when prompted to secure the identifiers themselves, Agent B might attempt to obfuscate the identifiers by converting sequential integers into Universally Unique Identifiers (UUIDs), mistakenly believing that cryptographic obscurity equates to structural authorization.[30]

Python

```python
# Agent B's Flawed ID Obfuscation Patch
import uuid

def create_user_profile(user_data):
    # The LLM generates a predictable UUIDv1 based on the current timestamp
    user_id = uuid.uuid1()
    db.insert("users", {"id": user_id, "data": user_data})
```

The Fallacy

Agent B's primary cognitive failure lies in consistently conflating authentication (verifying *who* the user is) with authorization (verifying *what* the user is permitted to access).[32] The LLM correctly applies the requireAuth middleware, ensuring that only users with a valid session token can access the route. However, when constructing the database query via the ORM, the LLM executes a direct lookup using only the primary key (id) extracted from the client-controlled request parameters.[33]

The model lacks the systemic, contextual awareness required to realize that multi-tenant applications demand compound WHERE clauses. The patch blindly trusts the client-provided id without intersecting it with the server-side state of the authenticated user (e.g., verifying that accountId matches req.user.id).[33] A prominent real-world example of this exact logic flaw involved a McDonald's API, where researchers decremented a sequential job application ID and exposed 64 million records containing personally identifiable information (PII) because the endpoint failed to intersect the requested ID with the session token.[35] Furthermore, when LLMs attempt to mitigate enumeration by substituting sequential IDs with UUIDs (specifically UUIDv1), they fail to recognize that time-based UUIDs are highly predictable. An attacker can create multiple test accounts, extract the UUIDv1 values, determine the exact timestamp generation pattern of the server, and predictably calculate the identifiers of newly created objects belonging to other users, rendering the obscurity defense entirely useless.[30] IDOR vulnerabilities are systemic logic flaws, and LLMs—which are highly optimized for localized syntax generation—fundamentally struggle to enforce the distributed context required for proper tenant isolation across application layers.[36]

Agent C Verification Matrix

Agent C must rely heavily on business logic fuzzing and compound AST validation to detect contextual authorization failures, as traditional static analysis often fails to differentiate between a legitimate read operation and an IDOR.[36]

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria

Static Analysis (AST)	Traverse the AST for ORM update, delete, or read methods (e.g., prisma.model.update, repository.delete, Model.findByIdAndUpdate). Inspect the properties passed within the where or condition objects.[33]	Reject any state-mutating or sensitive read database query where the conditional filter relies exclusively on a client-provided parameter (like id) without including a server-derived ownership claim (e.g., accountId: req.user.id or tenant_id).[33]
Static Analysis (AST)	Identify the implementation of UUID generation libraries in the context of database primary keys or access control patches.	Flag the usage of predictable UUID versions (e.g., uuidv1) and strictly enforce the usage of cryptographically secure random identifiers (e.g., uuidv4) to prevent identifier enumeration.[30]
Dynamic Validation	Execute a multi-actor, cross-tenant state manipulation sequence. Authenticate as User A and create an object. Authenticate as User B, capture the session token, and attempt to mutate User A's object by passing its identifier to the patched endpoint.[31]	If User B successfully alters the state, retrieves the data, or deletes User A's object, the IDOR persists. The patch is categorically rejected as a failure of contextual authorization.[31]

5. Race Conditions and Time-of-Check to Time-of-Use (TOCTOU)

The Vulnerability & The Exploit Context

Agent A exploits a temporal vulnerability in the application's sequential logic. By sending highly concurrent HTTP requests or manipulating symbolic links during a brief window of file processing, Agent A manages to exceed a strict business logic limit. This results in severe consequences, such as applying a single-use promotional code multiple times, withdrawing funds in excess of an account balance, or replacing a validated safe file with a malicious

symbolic link pointing to a protected system resource.[39]

Agent B's "Naive Patch" (The Mistake)

When tasked with resolving a concurrent race condition, Agent B invariably generates a sequential "check-then-act" control flow, operating under the assumption that standard procedural logic will prevent the exploit.

JavaScript

```javascript
// Agent B's Flawed File System TOCTOU Patch
const fs = require('fs');

function readUserFile(filename) {
  // Time of Check: The LLM verifies the file exists
  if (fs.existsSync(filename)) {
    // Time Gap: An attacker swaps the file for a symlink pointing to /etc/shadow
    // during this precise context switch

    // Time of Use: The LLM reads the file, unaware it has been modified
    return fs.readFileSync(filename, 'utf-8');
  }
  throw new Error('File not found');
}
```

For database operations, such as a financial withdrawal or inventory decrement, the LLM will structure the transaction similarly, relying on application-layer arithmetic:

JavaScript

```javascript
// Agent B's Flawed Database Race Condition Patch
const db = require('./database');

async function processWithdrawal(userId, withdrawalAmount) {
  // The LLM wraps the logic in a standard transaction
  await db.query('BEGIN');

  // Time of Check
```

```
  const user = await db.query('SELECT balance FROM accounts WHERE id =?', [userId]);

  if (user.balance >= withdrawalAmount) {
    // Time Gap: Concurrent requests pass the check simultaneously
    // Application layer arithmetic determines the new balance
    const newBalance = user.balance - withdrawalAmount;

    // Time of Use: The last transaction to commit overwrites the others
    await db.query('UPDATE accounts SET balance =? WHERE id =?',);
    await db.query('COMMIT');
    return true;
  }
  await db.query('ROLLBACK');
  return false;
}
```

The Fallacy

Agent B's logic reveals a profound disconnect between synchronous, single-threaded code generation and the realities of concurrent, multi-threaded execution environments.[41] The LLM structures the code linearly, operating under the fallacy that the state of the system will remain completely static between the condition evaluation (Time of Check) and the subsequent execution (Time of Use).[42] In the file system scenario, the LLM does not account for the operating system's ability to interject context switches. This provides an attacker with a microscopic window to replace a legitimate file with a malicious symbolic link.[40] The model is unaware that it must bypass the check altogether and leverage atomic OS-level operations—such as handling ENOENT natively during the read operation, or utilizing secure system calls like openat() which resolve file operations relative to a directory file descriptor, thereby eliminating the race window.[41]

In the database scenario, the LLM assumes that merely placing the check-then-act logic within a standard database transaction solves the race condition. It fundamentally misunderstands database isolation levels. The default isolation level for most database engines (like PostgreSQL's READ COMMITTED) allows concurrent transactions to read the exact same initial state before either transaction has committed an update.[45] Because the LLM performs the arithmetic in the application layer (const newBalance = user.balance - withdrawalAmount), multiple parallel requests read the same balance, pass the validation gate simultaneously, and clobber the subsequent updates. The final commit simply overwrites the previous ones, leading to a negative balance or an inventory limit overrun.[46] The LLM fails to implement strict row-level locking mechanisms (e.g., SELECT... FOR UPDATE) or rely on atomic database-layer decrements, allowing the race condition to persist despite the transaction boundary.[46]

Agent C Verification Matrix

Detecting TOCTOU and concurrency flaws requires Agent C to identify non-atomic sequences in static analysis and utilize aggressive connection pooling mechanisms during dynamic validation.

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria
Static Analysis (AST)	Traverse the AST for sequential pairings of file-checking APIs (e.g., existsSync, access, stat) followed immediately by file-consumption APIs (e.g., readFileSync, open) operating on the same variable.[41]	Flag the separation of check and use as a classic TOCTOU anti-pattern. Mandate the use of unified atomic operations with appropriate exception handling (e.g., try/catch blocks catching native fs errors).[41]
Static Analysis (AST)	Identify database read-modify-write patterns operating on numerical constraints (e.g., balances, inventory, usage quotas) within transaction blocks.	Reject patches that perform application-layer arithmetic on database fields without deploying explicit row-level locking (SELECT... FOR UPDATE) or atomic database-layer decrements (UPDATE table SET val = val - 1 WHERE val >= 1).[46]
Dynamic Validation	Execute an automated limit overrun race condition payload by utilizing tools like Burp Suite Repeater, Turbo Intruder, or custom Python scripts to send identical state-mutating requests simultaneously over a single multiplexed HTTP/2 or TCP connection.[39]	If the final state of the database reflects multiple successful operations exceeding the logical constraint (e.g., a balance drops below zero, or a single-use token is consumed twice), the patch fails to resolve the data race and is rejected.[39]

6. NoSQL and Command Injection

The Vulnerability & The Exploit Context

Agent A discovers that user input is being passed directly to a backend system interpreter without adequate sanitization. In one instance, injecting the payload {"$ne": null} into an authentication endpoint bypasses the password verification logic.[47] In another instance, appending ; cat /etc/passwd to a diagnostic tool input results in arbitrary operating system command execution, granting Agent A a remote shell on the host.[6]

Agent B's "Naive Patch" (The Mistake)

When attempting to patch NoSQL injection, Agent B frequently relies on superficial type coercion or entirely ignores the structural complexity of NoSQL document query drivers.

JavaScript

```javascript
// Agent B's Flawed NoSQL Injection Patch
const express = require('express');
const app = express();
app.use(express.json()); // Parses incoming JSON payloads

app.post('/login', async (req, res) => {
  // The LLM accepts a raw JSON object from the request body
  const { username, password } = req.body;

  // The LLM relies on the database driver to process the object natively
  const user = await db.collection('users').findOne({
    username: String(username), // Weak type casting
    password: password        // FATAL MISTAKE: Raw object injection
  });
});
```

For OS Command Injection, Agent B attempts to sanitize the input by explicitly stripping known dangerous characters, but it continues to invoke the system shell for the final execution phase:

Go

```go
// Agent B's Flawed Command Injection Patch
package main
```

```go
import (
    "os/exec"
    "strings"
)

func viewLog(filename string)byte {
    // The LLM relies on insufficient string replacement to sanitize input
    cleanName := strings.ReplaceAll(filename, ";", "")
    cleanName = strings.ReplaceAll(cleanName, "&", "")

    // FATAL MISTAKE: Passing input to a shell interpreter via a single string
    cmd := exec.Command("sh", "-c", "tail -n 10 /var/log/" + cleanName)
    output, _ := cmd.CombinedOutput()
    return output
}
```

The Fallacy

Agent B addresses injection flaws through the lens of traditional SQL injection, attempting to apply string manipulation and lexical escaping techniques to environments governed by entirely different execution models. In the context of NoSQL databases like MongoDB, queries are not constructed via string concatenation; they are built using structured BSON/JSON objects.[47] When the LLM accepts raw, nested data from req.body without strict schema validation, an attacker can supply an object containing structural query operators (e.g., {"username": "admin", "password": {"$ne": null}}).[47] The database engine parses the payload and evaluates the $ne (not equal) operator natively, completely bypassing the authentication logic because the password does not equal null. The LLM fails to recognize that NoSQL injection is an *operator* and *type-confusion* vulnerability, not a string escape problem.[50] Even when attempting to fix this, the LLM might overlook nested injections, such as the Mongoose populate().match vulnerabilities (CVE-2024-53900), where operators are smuggled deep within relational lookup parameters.[52]

Regarding OS Command Injection, the LLM exhibits a persistent, documented preference for shell invocation (sh -c or os.system) because it allows for the convenient, human-like chaining of commands, pipes, and environment variables.[53] The LLM attempts to sanitize the input by removing specific metacharacters (e.g., ; and &), leaving the application exposed to alternate command separators and evasion techniques (e.g., ||, backticks, newlines %0a, or inline execution via $(...)).[54] The model fails to grasp the fundamental security principle of structural isolation: bypassing the shell entirely and passing arguments directly as an array to the underlying operating system execution API (e.g., exec.Command("tail", "-n", "10", filename)). When arguments are passed in an array without a shell interpreter, shell metacharacters lose their special meaning and are treated purely as literal strings, rendering command injection

impossible.[53]

Agent C Verification Matrix

Agent C must enforce strict semantic boundaries, utilizing tools like Semgrep to detect shell invocations and dynamic payloads to fuzz operator boundaries.

Verification Type	Target Pattern / Execution Logic	Remediation Detection Criteria
Static Analysis (AST)	Traverse the AST for NoSQL database queries (e.g., collection.find, collection.findOne) that accept parameters directly derived from dynamic HTTP request bodies (req.body, req.query).[49]	Require strict schema validation logic (e.g., utilizing libraries like Joi or Zod) or enforce that the input is explicitly cast as a scalar primitive (e.g., typeof password === 'string') before it interacts with the database driver.[47]
Static Analysis (AST)	Identify CallExpression nodes responsible for process execution (e.g., exec.Command, child_process.exec, Runtime.getRuntime().exec) . Utilize Semgrep rules like run-shell-injection to trace tainted data flows.[53]	Categorically reject any patch where the command argument array initiates a shell interpreter (sh, bash, cmd.exe) while concatenating user-derived variables. Enforce binary array invocation.[53]
Dynamic Validation	Inject BSON/JSON operator payloads containing boolean evaluation operators ($ne, $gt, $regex, $where) into all parameters interacting with the NoSQL database layer.[49]	If the query logic is successfully manipulated or bypassed (e.g., the endpoint returns valid user records without an exact password match), the NoSQL defense is insufficient and rejected.[58]
Dynamic Validation	Fuzz the input parameters with diverse shell escape sequences including inline	Any observable system response indicating out-of-band execution

	execution $(id), pipeline operators |, backticks, and variable expansion %PATH%.[54]	(e.g., DNS resolution to a collaborator server) or command chaining results in immediate patch rejection.[48]

Synthesis and Recommendations

The deployment of autonomous multi-agent swarms for software remediation offers unprecedented scalability, yet it simultaneously introduces profound systemic risks into the software development lifecycle. As evidenced by this taxonomy, LLM-driven engineers (Agent B) consistently exhibit a "Format-Reliability Gap," prioritizing functional syntax and rapid execution over structural security guarantees.[6] Their reasoning is constrained by the prevailing patterns in their training data, leading them to rely on naive string sanitization, permissive wildcard configurations, implicit environment parsing, and non-atomic procedural flows.[9]

These models inherently struggle with abstract security concepts such as contextual authorization, temporal execution gaps, and the adversarial implications of the environments they manipulate (e.g., CI/CD pipelines and Docker isolation mechanisms).[11] Consequently, the efficacy and safety of an autonomous remediation swarm rely entirely on the rigor of the auditing component (Agent C).

By deploying a verification matrix grounded in strict Abstract Syntax Tree (AST) pattern matching, inter-procedural data-flow analysis, and aggressive dynamic payload injection, organizations can bridge the cognitive blindspots of LLM code generation. Implementing methodologies such as the Agent Robustness Evaluation Protocol (AREP) ensures that patches are evaluated not just for functional correctness, but for their resilience against secondary exploitation.[4] This defense-in-depth architecture ensures that the velocity of AI-driven vulnerability patching is matched by the structural guarantees necessary to secure modern digital infrastructure.

Works cited

1. When AI Fails, What Works? A Data-Driven Taxonomy of Real-World Risk Mitigation Strategies - arXiv, accessed April 23, 2026, https://arxiv.org/html/2603.04259v1
2. Building Autonomous, Responsible and Intelligent Agentic AI Systems - Infosys, accessed April 23, 2026, https://www.infosys.com/techcompass/documents/building-autonomous-aria-systems.pdf
3. tmgthb/Autonomous-Agents: Autonomous Agents (LLMs) research papers. Updated Daily. - GitHub, accessed April 23, 2026, https://github.com/tmgthb/Autonomous-Agents
4. Why AI Agents Fail: A Taxonomy of Failure Modes in Autonomous LLM-Based

Systems, accessed April 23, 2026, https://www.researchgate.net/publication/404001571_Why_AI_Agents_Fail_A_Taxonomy_of_Failure_Modes_in_Autonomous_LLM-Based_Systems
5. A Systematic Study of LLM-Based Architectures for Automated Patching - arXiv, accessed April 23, 2026, https://arxiv.org/html/2603.01257v1
6. Surgical Repair of Insecure Code Generation in LLMs From Mechanistic Diagnosis to Deployment-Ready Intervention - arXiv, accessed April 23, 2026, https://arxiv.org/html/2604.16697v1
7. The Most Common Security Vulnerabilities in AI-Generated Code | Blog - Endor Labs, accessed April 23, 2026, https://www.endorlabs.com/learn/the-most-common-security-vulnerabilities-in-ai-generated-code
8. The Hidden Risks of LLM-Generated Web Application Code: A Security-Centric Evaluation of Code Generation Capabilities in Large Language Models - arXiv, accessed April 23, 2026, https://arxiv.org/html/2504.20612v1?ref=canartuc.com
9. From .env to Leakage: Mishandling of Secrets by Coding Agents - Knostic, accessed April 23, 2026, https://www.knostic.ai/blog/claude-cursor-env-file-secret-leakage
10. Adversarial Bug Reports as a Security Risk in Language Model-Based Automated Program Repair - arXiv, accessed April 23, 2026, https://arxiv.org/html/2509.05372v1
11. Adversarial Bug Reports as a Security Risk in Language Model-Based Automated Program Repair - arXiv, accessed April 23, 2026, https://arxiv.org/html/2509.05372v2
12. How to fix Docker: Permission denied - Stack Overflow, accessed April 23, 2026, https://stackoverflow.com/questions/48957195/how-to-fix-docker-permission-denied
13. Your AI Wrote a CORS Config That Lets Any Website Read Your API - DEV Community, accessed April 23, 2026, https://dev.to/busyagents/your-ai-wrote-a-cors-config-that-lets-any-website-read-your-api-1h79
14. Access-Control-Allow-Origin header - HTTP - MDN Web Docs, accessed April 23, 2026, https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/Access-Control-Allow-Origin
15. CORS: Cannot use wildcard in Access-Control-Allow-Origin when credentials flag is true, accessed April 23, 2026, https://stackoverflow.com/questions/19743396/cors-cannot-use-wildcard-in-access-control-allow-origin-when-credentials-flag-i
16. How to Fix "No 'Access-Control-Allow-Origin' Header Present" - StackHawk, accessed April 23, 2026, https://www.stackhawk.com/blog/fixing-no-access-control-allow-origin-header-present/
17. Cannot get VLLM Docker to launch - memory errors. : r/LocalLLM - Reddit, accessed April 23, 2026,

https://www.reddit.com/r/LocalLLM/comments/1skuzsc/cannot_get_vllm_docker_to_launch_memory_errors/
18. Agent faces permission errors - Friends of the Crustacean - Answer Overflow, accessed April 23, 2026, https://www.answeroverflow.com/m/1478139511107096747
19. The Strange Case of How We Escaped the Docker Default Container - CyberArk, accessed April 23, 2026, https://www.cyberark.com/resources/threat-research-blog/the-strange-case-of-how-we-escaped-the-docker-default-container
20. LLM Zero-Day Discovery: Why Container Isolation Fails - Edera, accessed April 23, 2026, https://edera.dev/stories/the-price-of-a-zero-day-vulnerability-is-an-api-call
21. Container Escape: New Vulnerabilities Affecting Docker and RunC - Palo Alto Networks, accessed April 23, 2026, https://www.paloaltonetworks.com/blog/cloud-security/leaky-vessels-vulnerabilities-container-escape/
22. CORS Implementation & Various Headers, CORS series (Part 2) | by Vipul Jain | Medium, accessed April 23, 2026, https://medium.com/@vipulparveenjain/cors-implementation-various-headers-cors-series-part-2-895a04ef481d
23. LLM05:2023 - SSRF Vulnerabilities, accessed April 23, 2026, https://owasp.org/www-project-top-10-for-large-language-model-applications/Archive/0_1_vulns/SSRF.html
24. Server-Side Request Forgery (SSRF) | Common Attacks & Risks - Imperva, accessed April 23, 2026, https://www.imperva.com/learn/application-security/server-side-request-forgery-ssrf/
25. Server-Side Request Forgery (SSRF) via MetaIssuer Regex Bypass ..., accessed April 23, 2026, https://github.com/sigstore/fulcio/security/advisories/GHSA-59jp-pj84-45mr
26. Server-Side Request Forgery (SSRF) - CVEs - page 4 - Feedly, accessed April 23, 2026, https://feedly.com/cve/cwe/918?page=4
27. SSRF Prevention in Node.js - OWASP Foundation, accessed April 23, 2026, https://owasp.org/www-community/pages/controls/SSRF_Prevention_in_Nodejs
28. Formal Verification for JavaScript Regular Expressions: a Proven Semantics and its Applications (Extended Version) - arXiv, accessed April 23, 2026, https://arxiv.org/pdf/2507.13091
29. From . in regex to SSRF — part 1 - Marcin Niemiec, accessed April 23, 2026, https://xvnpw.medium.com/from-in-regex-to-ssrf-part-1-31d5706854ef
30. IDOR Vulnerabilities Explained: A Researcher's Guide to Authorization Flaws - Medium, accessed April 23, 2026, https://medium.com/@ud4y25/idor-vulnerabilities-explained-a-researchers-guide-to-authorization-flaws-82030def0e28
31. Insecure Direct Object Reference (IDOR) Attack Guide - Hackviser, accessed April 23, 2026, https://hackviser.com/tactics/pentesting/web/idor

32. Insecure Direct Object References (IDOR) - Invicti, accessed April 23, 2026, https://www.invicti.com/learn/insecure-direct-object-references-idor
33. Multiple IDOR in message and tag routers - missing accountId ..., accessed April 23, 2026, https://github.com/blinkospace/blinko/issues/1112
34. AI Gone Rogue: Why Authorization, Not Instructions, Keeps LLMs in Check - Oso, accessed April 23, 2026, https://www.osohq.com/post/why-authorization-keeps-llms-in-check
35. Authorization Bugs Are Having Their SQL Injection Moment - ZeroPath Blog, accessed April 23, 2026, https://zeropath.com/blog/idor-crisis-2025
36. Catching IDORs, Broken Authorization, and Other Logic Issues with Semgrep AI-Powered Detection, accessed April 23, 2026, https://semgrep.dev/blog/2025/ai-powered-detection-with-semgrep/
37. IDOR Vulnerability Explained: Why Insecure Direct Object References Persist, accessed April 23, 2026, https://www.aikido.dev/blog/idor-vulnerability-explained
38. From detection to prevention: How Zen stops IDOR vulnerabilities at runtime, accessed April 23, 2026, https://www.aikido.dev/blog/zen-stops-idor-vulnerabilities
39. Race conditions | Web Security Academy - PortSwigger, accessed April 23, 2026, https://portswigger.net/web-security/race-conditions
40. TOCTOU race condition allows symlink attacks during lock file creation - GitHub, accessed April 23, 2026, https://github.com/tox-dev/filelock/security/advisories/GHSA-w853-jp5j-5j7f
41. Understanding TOCTOU: The Race Condition Hiding in Your Code - DEV Community, accessed April 23, 2026, https://dev.to/codewithveek/understanding-toctou-the-race-condition-hiding-in-your-code-43nh
42. Guide to Identifying and Exploiting TOCTOU Race Conditions in Web Applications, accessed April 23, 2026, https://fdzdev.medium.com/guide-to-identifying-and-exploiting-toctou-race-conditions-in-web-applications-c5f233e32b7f
43. Winning the race against TOCTOU vulnerabilities in C & C++ - Sonar, accessed April 23, 2026, https://www.sonarsource.com/blog/winning-the-race-against-toctou-vulnerabilities
44. How to manipulate the execution flow of TOCTOU attacks - Olivia A. Gallucci, accessed April 23, 2026, https://oliviagallucci.com/how-to-manipulate-the-execution-flow-of-toctou-attacks/
45. A Race to the Bottom - Database Transactions Undermining Your AppSec - Doyensec's Blog, accessed April 23, 2026, https://blog.doyensec.com/2024/07/11/database-race-conditions.html
46. Race Conditions in REST APIs: A Developer's Guide to Building Reliable Systems - Medium, accessed April 23, 2026, https://medium.com/@mgaurang123/race-conditions-in-rest-apis-a-developers-guide-to-building-reliable-systems-42d4f8eabc1e

47. Preventing SQL/NoSQL Injection Attacks in MongoDB - DataCamp, accessed April 23, 2026, https://www.datacamp.com/tutorial/preventing-sql-no-sql-injection-attacks-in-mongo-db
48. Command Injection - OWASP Foundation, accessed April 23, 2026, https://owasp.org/www-community/attacks/Command_Injection
49. NoSQL Injection Explained: What It Is and How to Prevent It - Bright Security, accessed April 23, 2026, https://brightsec.com/blog/nosql-injection-explained-what-it-is-and-how-to-prevent-it/
50. NoSQL injection | Web Security Academy - PortSwigger, accessed April 23, 2026, https://portswigger.net/web-security/nosql-injection
51. JavaScript type confusion: Bypassed input validation (and how to remediate) - Snyk, accessed April 23, 2026, https://snyk.io/blog/remediate-javascript-type-confusion-bypassed-input-validation/
52. NoSQL injection - HackTricks, accessed April 23, 2026, https://hacktricks.wiki/en/pentesting-web/nosql-injection.html
53. Avoid command injection - Datadog Docs, accessed April 23, 2026, https://docs.datadoghq.com/security/code_security/static_analysis/static_analysis_rules/go-security/command-injection/
54. Shell Injection | DeepTeam by Confident AI - The LLM Red Teaming Framework, accessed April 23, 2026, https://trydeepteam.com/docs/red-teaming-vulnerabilities-shell-injection
55. Security: Unvalidated User Input in Shell Command Arguments · Issue #177 · kardolus/chatgpt-cli - GitHub, accessed April 23, 2026, https://github.com/kardolus/chatgpt-cli/issues/177
56. Command injection detection | Moderne Docs, accessed April 23, 2026, https://docs.moderne.io/openrewrite-advanced-program-analysis/security/command-injection/
57. Security Companies Under Attack: Defense in the Agentic Era | Semgrep, accessed April 23, 2026, https://semgrep.dev/blog/2026/security-companies-under-attack-defense-in-the-agentic-era/
58. Exploiting NoSQL Operator Injection to Bypass Authentication - YouTube, accessed April 23, 2026, https://www.youtube.com/watch?v=DBNmAJaWcGk